TABLEAU DATA ANALYST - ADVANCED PRACTICE EXAM

ANSWER KEY WITH EXPLANATIONS

QUICK ANSWER KEY

Q1: B	Q11: D	Q21: B	Q31: D	Q41: B	Q51: A
Q2: C	Q12: C	Q22: D	Q32: B	Q42: C	Q52: B
Q3: A	Q13: B, C, D	Q23: B	Q33: B	Q43: A, D	Q53: C
Q4: B	Q14: C	Q24: C	Q34: C	Q44: D	Q54: D
Q5: B	Q15: D	Q25: A	Q35: A, C	Q45: B	Q55: C
Q6: C	Q16: C	Q26: C	Q36: C	Q46: B	Q56: A
Q7: D	Q17: C	Q27: D	Q37: A, C, D	Q47: C	Q57: B
Q8: B	Q18: D	Q28: D	Q38: C-F-A-D-E-G-B	Q48: A, C, D	Q58: C
Q9: B, D	Q19: C	Q29: B	Q39: B-C-A-D	Q49: C	Q59: D
Q10: B-D-C-A	Q20: D	Q30: B, C	Q40: B-D-C-A-E-F	Q50: D	Q60: C

Scoring: give yourself 1 point per question only when the answer is COMPLETELY right (all letters on multi-selects; the exact sequence on build lists). 39+/60 keeps you on track for the real exam's 65% pass mark; 45+ means exam-ready.

DETAILED EXPLANATIONS

Question 1 - Correct answer: B

Initial SQL runs a command once, at the moment the connection opens (per session). It is the documented mechanism for session setup - temporary tables, session variables, or security context - and every query Tableau then sends can reference #UserContext. Custom SQL instead defines the SELECT used to fetch data, and filters operate on rows, not on sessions.

Question 2 - Correct answer: C

An aggregated extract physically rolls rows up to the visible dimensions and rolls dates up to the chosen level. Only YEAR-level date information exists in the .hyper file, so no worksheet can drill to month. An incremental refresh appends rows at the stored grain - it cannot restore detail. Re-creating the extract without the roll-up is the only fix.

Question 3 - Correct answer: A

Tableau joins follow SQL semantics: NULL never equals anything, including another NULL. Rows with NULL keys fail the join condition, so the 500 NULL-email customers remain (left join keeps them) with NULL preference fields, and they can never pair with the 120 NULL rows. Matching them would require a calculated key such as IFNULL([Email], 'none') on both sides.

Question 4 - Correct answer: B

Unions align columns by exact field name only. CustID and CustomerID arrive as two separate columns, each NULL on the rows that came from the other file. Rename one side to match the other (before the union, or in a Clean step) so the two columns merge.

Question 5 - Correct answer: B

A blend aggregates the secondary source to the level of the linking dimensions present in the view. The view links on Region only, so East's three monthly target rows all collapse onto the East mark and SUM([Target]) = 10,000 + 10,000 + 10,000 = 30,000. Add Month to the view and the blend links at Region + Month, showing 10,000.

Question 6 - Correct answer: C

Custom SQL can embed a parameter in the SELECT (e.g., WHERE Region = <Parameters.Region>). When the user changes the parameter, Tableau re-issues the query with the new value, so the database itself returns only the requested rows. Data source filters and quick filters cannot inject a parameter's value into the SQL a live connection sends.

Question 7 - Correct answer: D

.tdsx is the packaged data source file: connection metadata plus the .hyper extract in one portable file - exactly what offline work needs. .tds carries only connection info (no data), .twb is a workbook without data, and .tfl is a Tableau Prep flow.

Question 8 - Correct answer: B

Data sampling in Prep (Quick select / Random / Stratified) only limits what the profile and preview panes display so the tool stays responsive. When the flow runs, it processes the entire input - all 5 million rows reach the output.

Question 9 - Correct answer: B, D

Hiding unused fields keeps them out of the metadata Tableau sends with every query, and extracts (the columnar Hyper engine) are the standard fix for large, slow live sources. Dozens of quick filters add query complexity, and disabling aggregation forces the view to render every row - both hurt performance.

Question 10 - Correct answer: B-D-C-A

Connect first. Clean with the Data Interpreter BEFORE reshaping - otherwise the title, subtotals, and footnotes become bogus rows in the pivot. Then pivot the twelve month columns into Month/Revenue rows, and finally rename the generated fields and build the line chart.

Question 11 - Correct answer: D

Relationship clauses pair fields of compatible types - a date cannot pair with an integer. Create YEAR([Order Date]) as a calculated field in the Orders logical table and define the relationship between it and [Budget Year] (keeping Region as a second pair). Relationships support calculated fields and multiple field pairs.

Question 12 - Correct answer: C

A context filter is applied before all other filters; Tableau materializes its result once (a temporary table / cached subset), and every subsequent query for that sheet runs against the much smaller set - eliminating the repeated full-table scans. Sorting and table-calculation filters do not reduce database work.

Question 13 - Correct answer: B, C, D

Double-clicking a logical table opens its physical (join) canvas, where you can join or union more tables at that grain. Because related tables stay separate, each table's measure is computed at its own level of detail and the aggregated results are combined at the viz's grain - no fan-out duplication of SUM. Relationship clauses CAN be edited and can use calculated fields, so option A is false.

Question 14 - Correct answer: C

A spatial join is built in the physical layer: join the two tables and set the join clause to the Intersects operator between the polygon geometry and the point (build points from lat/long with MAKEPOINT, or use existing geometry). Each store row then inherits the attributes of the territory polygon that contains it.

Question 15 - Correct answer: D - 150

EXCLUDE removes [Region] from the calculation's level, leaving Customer: Anna 50, Ben 70, Cara 60 + 90 = 150, Dana 20. On the West | Cara mark the value is Cara's total across both regions: 150. (Plain SUM would show 90 on that mark; {FIXED : SUM} would show the grand total 290 on every row.)

Question 16 - Correct answer: C - 60

The FIXED LOD builds one value per (Region, Customer): East-Cara = 60 and West-Cara = 90. The dimension filter cannot change those FIXED values, but it DOES remove the West rows from the view. Only the East-Cara bucket survives on Cara's mark, so AVG(M) = 60. Without the filter it would average both buckets: AVG(60, 90) = 75. This two-layer interplay (LOD ignores the filter; the view still filters its rows) is the trap.

Question 17 - Correct answer: C - returns 3

The inner FIXED LOD fetches each customer's first order date; DATEDIFF('year', first, this) counts crossed year boundaries: 2022, 2023, 2024 -> 3. A reverses the arguments (-3); B uses MAX (the latest order); D mixes an aggregate MIN([Order Date]) with a row-level field outside any LOD, throwing the 'cannot mix aggregate and non-aggregate' error.

Question 18 - Correct answer: D - 100

Read nested LODs inside-out. Inner: one SUM per (Region, Customer) - West: 50, 100, 150. Outer: for each Region, AVG of those customer totals: (50 + 100 + 150) / 3 = 100.

Question 19 - Correct answer: C - INDEX() = 5, FIRST() = -4

INDEX() counts positions from the first row of the partition, starting at 1, so the 5th row = 5. FIRST() returns the OFFSET from the first row (first row = 0), so the 5th row is 4 rows back = -4.

Question 20 - Correct answer: D - 30

Offsets (-1, 1) center the window: previous, current, next = (20 + 30 + 40) / 3 = 30. A trailing window (-2, 0) would give the answer for a different specification.

Question 21 - Correct answer: B - 100

FIRST() is evaluated per mark: on the 3rd mark it equals -2, so LOOKUP(SUM([Sales]), FIRST()) reaches back 2 marks to January = 100. Every mark in the view displays the first month's value - the classic 'index to the first row' pattern.

Question 22 - Correct answer: D - RANK_MODIFIED

Modified competition ranking gives tied items the HIGHEST of the tied positions: 500 -> 1, the two 400s -> 3, 300 -> 4, i.e., 1, 3, 3, 4. RANK gives 1, 2, 2, 4; RANK_DENSE gives 1, 2, 2, 3; RANK_UNIQUE gives 1, 2, 3, 4.

Question 23 - Correct answer: B - #2024-02-29#

DATEADD('month') targets the same day-of-month where it exists; January 31 has no counterpart in February, so Tableau clamps to the month's last day - February 29 in the leap year 2024.

Question 24 - Correct answer: C - 1

DATEDIFF counts unit boundaries crossed, not elapsed time. Jan 31 -> Feb 1 crosses exactly one month boundary, so the answer is 1 - even though only one day elapsed.

Question 25 - Correct answer: A

SPLIT's token index can be negative, counting from the END: -1 is the last token, so US-2024-1186 -> 1186. RIGHT([Code], 4) only happens to work while the number is four characters long, and token 3 only works if the token count never changes.

Question 26 - Correct answer: C - 4600

ROUND's digits argument may be negative: -2 rounds to hundreds, so 4567.89 -> 4600.

Question 27 - Correct answer: D - FLOOR

FLOOR always rounds toward negative infinity: FLOOR(-3.2) = -4. CEILING(-3.2) = -3, INT(-3.2) = -3 (INT truncates toward zero), and ROUND(-3.2, 0) = -3.

Question 28 - Correct answer: D - "On time"

A comparison with NULL is never TRUE; Tableau's IF treats it as not-true and falls through to the ELSE branch, returning 'On time'. (With no ELSE the result would be NULL.) To handle missing dates explicitly, branch on ISNULL([Ship Date]) first.

Question 29 - Correct answer: B

AVG([Sales]) averages the row-level sales WITHIN each day's mark, and {FIXED : AVG([Sales])} averages all underlying rows - neither is the average of the daily totals. WINDOW_AVG(SUM([Sales]), FIRST(), LAST()) averages the aggregated marks across the whole partition - exactly the average daily total.

Question 30 - Correct answer: B, C

Parameter lists sourced from a field can refresh when the workbook opens (dynamic parameters), and when a parameter action fires on a multi-mark selection, Tableau can assign an aggregated value (ATTR / SUM / AVG / MIN / MAX) of the selected marks' field. A parameter holds exactly ONE value at a time, and changing it never triggers a refresh.

Question 31 - Correct answer: D

By default, Tableau computes grand totals by re-running the measure's aggregation over ALL underlying rows at the total level - AVG across the entire data set - not the average of the displayed regional averages. That is why totals on averaged measures often look 'off' compared to eyeballing the column.

Question 32 - Correct answer: B

DISTANCE takes two point geometries plus a unit literal - 'meters', 'km', or 'miles'. MAKEPOINT(lat, lon) builds each point. The call is invalid without the unit argument, and DISTANCE never takes four scalar coordinates.

Question 33 - Correct answer: B - #2024-10-13#

DATETRUNC('week') rolls a date back to the start of its week. With Tableau's default date properties (weeks start Sunday), Wednesday Oct 16 truncates to Sunday Oct 13. A workbook with a different start-of-week setting would truncate differently.

Question 34 - Correct answer: C - 2

INCLUDE adds the finer [Order Date] beneath the month-level view, so COUNTD is computed per day: Jan 1 = 2, Jan 2 = 2, Jan 3 = 1. The view then aggregates those values with MAX -> 2. (SUM would give 5; AVG about 1.7.)

Question 35 - Correct answer: A, C

COUNT counts only non-null rows (COUNTD also ignores nulls - it does NOT count NULL as a member). If every row in a group is null, SUM returns NULL, not 0 - wrap with ZN when zero is required. MEDIAN is not supported by every live connection (extracts/files compute it fine), so D is false.

Question 36 - Correct answer: C - "*"

ATTR returns the field's value only when it is unique within the mark; when several values occur it returns the special string '*'. Add Region to the level of detail, or use MIN/MAX, to surface a real value.

Question 37 - Correct answer: A, C, D

DATETRUNC rolls a date back to the first of the month; NOW() returns a datetime while TODAY() returns a date; DATEADD accepts negative offsets. DATENAME returns a STRING ('Wednesday'), never an integer - DATEPART is the numeric version.

Question 38 - Correct answer: C-F-A-D-E-G-B

Extract filters -> data source filters -> context filters -> LOD expressions computed -> dimension filters -> measure filters -> table calculation filters. FIXED LODs are immune to dimension and measure filters but honor the three earlier stages; table calculations always run last, on the aggregated results.

Question 39 - Correct answer: B-C-A-D

First compute the first order date per customer (the FIXED LOD), then convert it to a year with a YEAR() wrapper to build a usable dimension, then place the cohort on the view with the measure, and finally validate that each customer's earliest order lands in the right cohort.

Question 40 - Correct answer: B-D-C-A-E-F

The sheets must exist on the dashboard first; then open Dashboard -> Actions and add a Filter action; choose the source and target sheets; set Run action on: Hover; add the targeted Region field so nothing else is filtered; then test the hover behavior.

Question 41 - Correct answer: B

Size: Automatic lets the dashboard canvas track the window, and Fit: Entire View scales the content to fill the canvas in both directions. Fit Width/Height fit only one dimension, and Fit: Normal keeps the content at its natural size (scrolling).

Question 42 - Correct answer: C

In the tooltip editor, use Insert -> Sheets -> choose the trend sheet; the insert dialog offers a size setting and a Filter option - enable it (All fields, or the targeted Customer field) so the embedded viz filters itself to the hovered mark. Hiding axes or highlight actions never filter a tooltip viz.

Question 43 - Correct answer: A, D

Floating items live on an absolute layer: they can overlap each other and the tiled layer, and a floating container can host sheets and objects like any container. Floating and tiled layouts can be freely mixed on one dashboard.

Question 44 - Correct answer: D

Right-click the specific mark -> Annotate -> Mark. A mark annotation attaches to that data point (moving with it and able to quote its values). Point annotations pin to a spot on the viz; Area annotations shade a rectangular region.

Question 45 - Correct answer: B

Edit the worksheet title, position the cursor where the value should appear, then Insert -> Parameters -> Region Selector. The title renders the parameter's CURRENT value and updates whenever the user changes the selection.

Question 46 - Correct answer: B

A Diverging palette with the Center fixed at 0 (Edit Colors -> Advanced) pins the red/blue boundary to zero. Sequential palettes - with Full Color Range or Stepped Color - have no midpoint, so small positives could still shade toward red.

Question 47 - Correct answer: C

Right-click the axis -> Format -> Scale (Numbers) and set the display units to Millions: 1,500,000 renders as 1.5M. No calculation and no data change, and the underlying values stay precise.

Question 48 - Correct answer: A, C, D

Dashboards can host web page objects (live URLs), download objects (one-click image/PDF/PPT export), and extension objects. There is no 'live SQL table' dashboard object - data lives in the data source pane.

Question 49 - Correct answer: C

A stacked area chart plots the month trend with Region stacked on Color, showing each region's magnitude AND the changing composition over time. (A 100%-stacked area is the variant when share-of-total matters more than magnitude.) Pies cannot show change over time.

Question 50 - Correct answer: D

On the Marks card, open Label, choose to allow some labels, and enable Min/Max: only the line's extreme points are labeled. Dragging the measure to Label labels every mark.

Question 51 - Correct answer: A

Right-click the axis -> Edit Axis -> set the Range to Fixed with start 0. Bar charts encode value by bar LENGTH, so a truncated axis distorts comparisons; fixing the range restores the honest baseline.

Question 52 - Correct answer: B

The Format menu's Lines pane (or Format -> Lines) controls Row and Column gridlines - set them to None. Shading/banding is removed via Format -> Shading. Clear Sheet removes pills, not formatting.

Question 53 - Correct answer: C

Set actions offer three running modes: Assign values to set (replaces membership each click), Remove from set, and Add to set - which grows the selection click by click, exactly what the scenario requires.

Question 54 - Correct answer: D

'Show all values' restores the target to its unfiltered state when the selection is cleared - right for exploratory filters. 'Exclude all values' empties the target, and 'Leave the filtered view' keeps the last filter stuck on.

Question 55 - Correct answer: C

Packed bubbles size circles by a measure with no nested hierarchy - the quickest spatial magnitude comparison. Treemaps add hierarchy plus two encodings; highlight tables encode color in a grid.

Question 56 - Correct answer: A

Certification is the governance badge: a steward marks the governed source as Certified and a gold badge appears next to it everywhere in Server/Cloud, signaling the officially endorsed source.

Question 57 - Correct answer: B

Setting a data quality warning (stale / freshness / deprecated) displays a visible warning badge on the data source and on content that uses it, so analysts immediately know the data is not fresh.

Question 58 - Correct answer: C

Data-driven alerts attach to a measure AXIS: open the view, right-click the numeric axis, and set the threshold. A pure text table has no continuous axis to attach the alert to - add an axis (bar/line) first.

Question 59 - Correct answer: D

The site role is the ceiling on capabilities: a Viewer can only interact, so granting a Save capability at project level cannot lift her past the role's maximum. She must be assigned Explorer (can publish) or Creator to save content.

Question 60 - Correct answer: C

Locking the permissions to the project stops inheritance from the parent project and forces all content and child projects inside to use exactly the permissions defined at that project - the standard tool for centrally managed, consistent access.

Study loop for the weak spots: for every LOD or calculation you missed, re-derive the result by hand with a small table before re-reading the explanation - the exam tests exactly this mental trace under time pressure.